



Validation of the Romanian Version of Autism Spectrum Quotient (AQ) and Empathy Quotient (EQ) in the General Population

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Abstract

Introduction The diagnosis of Autism Spectrum Disorder (ASD) can be difficult to achieve, especially since the diagnostic criteria have changed in the last decade. This challenge is even greater for Romanian clinicians, as there are currently no screening tools validated for diagnosing ASD in adults within the Romanian population.

Purpose This study aims to validate two screening instruments used to identify autistic traits in the Romanian general population: the Autism Spectrum Quotient (AQ) and the Empathy Quotient (EQ).

Methods Data was collected by administering the Romanian version of the scales to a sample of 916 Romanian native speakers from the non-clinical population. 108 individuals were re-tested after 2 years. The scales were validated by determining their reliability with respect to the stability of the results and evaluating their construct and structural validity in correlation with the following instruments: QCAE (Questionnaire of Cognitive and Affective Empathy), PAQ (Perth Alexithymia Questionnaire) and BVAQ (Bermond-Vorst Alexithymia Questionnaire).

Results Our data revealed that, regarding reliability, the Romanian version of AQ presents slightly lower internal consistency compared to the original validation study. Concerning structural validity, confirmatory factor analysis (CFA) revealed suboptimal fit indices for the original 5-factor structure. Based on a subsequent exploratory factor analysis (EFA), a 2-factor solution returned improved fit indices.

Conclusion Despite employing more items to measure the constructs than needed, this study confirmed that both instruments present a valuable resource for clinicians working with neurodivergent patients, as interest in this area of pathology has increased greatly in Romania over the past few years. Practical implications and methodological issues regarding the two instruments are discussed.

Keywords Autism spectrum disorder · Validation study · Autism quotient · Empathy quotient · Romania

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In Romania, Autism Spectrum Disorder (ASD) is diagnosed only through clinical criteria because there are no validated screening tools for the local population. The situation is further complicated by the fact that the Romanian public healthcare system still uses the ICD-10-AM (2002) coding system [1], and the National Health Insurance House considers all autism-related conditions, including Asperger's Syndrome (AS), as pertaining to childhood only. Consequently, children with ASD receive medical care covered by insurance until they turn 18, after which they lose all their benefits upon transitioning to adult services. To ensure their benefits are retained after the age of 18, their diagnosis is changed to other conditions that clinicians consider related to autism, such as cluster A or C personality disorders or schizophrenia, as well as to other pathologies that are comorbidities of ASD, like affective disorders or obsessive-compulsive disorder, making the identification of autism challenging. Furthermore, the stigma associated with mental health issues causes many parents to be reluctant to seek help from psychiatrists [2], which leads many people to reach adulthood without an official diagnosis. Upon receiving a diagnosis of ASD, the conventional recommendation is to enroll these children in specialized educational institutions. However, these schools often encounter criticism regarding the quality of treatment and the adequacy of care provided (Crețu, A., 2015, unpublished PhD thesis). As children with undiagnosed ASD transition into adulthood, they become harder to diagnose due to the outdated coding system and lack of screening tools. Thus, very few adult patients are newly diagnosed despite the high demand, which has led many of them to seek psychiatric help in other countries.

There are no studies on the prevalence of ASD among adults in Romania, and cases often go unnoticed due to the changes in diagnosis when individuals are transferred to adult services. The only existing study cites a 14.3% prevalence of ASD in a Romanian cohort of children aged 7 to 8, which includes both children from the general population and those with special educational needs [3]. Consequently, this figure is significantly higher than those reported in Europe, where the prevalence rates for ASD in children range from 0.38% in Germany to 1.55% in Spain [4], with a global prevalence of 1% (0.24–2.68 for Europe) [5]. The prevalence of AS in the general population is estimated to be between 0.2% and 0.7% [6].

The introduction of DSM-5 in 2013 corrected the diagnostic incongruence between children and adults by introducing ASD and allowing autism to be recognized in both demographics [7], thereby ensuring continuity of care without the need for coding changes. This resulted in the unofficial use of DSM-5 coding by Romanian psychiatrists, who reference it in medical papers, even though the public healthcare system does not officially recognize it.

The issue of screening and diagnosing ASD in adults is a relatively new endeavor, preparing the way for the adult diagnostic mentioned by the DSM-5. The Adult Asperger Assessment battery (AAA), developed by Baron-Cohen and his team [8], is a comprehensive instrument that includes autism spectrum testing (Autism Spectrum Quotient– AQ), emotions and empathy testing (Empathy Quotient– EQ). A Relative Quotient (RQ) questionnaire, preferably completed by the main childhood caregiver of the patient or by a former educator, provides insight into the patient's childhood history and behavior patterns [8, 9, 10, 11] and is sometimes added to the AAA battery when more history data is needed. The AAA battery was selected for this study due to its widespread availability in multiple languages and at no cost, with the hope that Romanian specialists will adopt it as a screening tool.

The AAA is a tool for screening and diagnosing autism without intellectual disability, initially referred to as High Functioning Autism (HFA) in the scale validation article [8, 10, 12]. Although DSM-5 includes AS in the autism spectrum and does not list HFA as a separate diagnosis, the authors of the AAA diagnosis instrument consider that AS and HFA, although clinically similar, have a significant difference related to development. HFA includes the autism spectrum elements (communication difficulties, low imagination, low attention mobility, and weak social abilities, all associated with increased adherence to routine) in individuals with cognitive and/or language delays in childhood [9, 11], defined as the inability to speak in separate words by the age of 2 and/or the inability to form short and simple sentences by the age of 3 [10]. AS does not include cognitive and/or language delays but has the same clinical characteristics as HFA. Both demand an average or over-average IQ level [13, 14]. There is an ongoing debate regarding the different terminologies used to describe autism in adults without intellectual or language delay. The DSM-5 dimensional approach was preferred in this paper, including AS and HFA under the ASD umbrella.

The AQ scale of the AAA battery is concordant with the DSM-IV-TR criteria [15], recording the autism spectrum elements in a 50-item self-report questionnaire analyzing five functional characteristics of autism without intellectual disability, each evaluated with 10 items: social abilities impairment; attention disturbance (flexibility, mobility, focus); attention to details; communication impairment; and poor imagination [8]. Scoring is made on a 1–4 Likert scale (1 - total agreement, 4 - total disagreement), with the subject receiving one point on each item if an abnormal score is recorded. To avoid bias, about half of the items were negatively formulated, the other half being positively formulated, corresponding to typical responses for people with a diagnosis of autism without intellectual disability. The AQ score varies between 0 and 50 [8, 16].

The EQ scale consists of 60 items, 40 analyzing empathy and 20 fillers with a score varying between 0 and 80 scored on the same Likert scale as the AQ, with 1 point for average empathy and 2 points for high empathy level [8]. Adults diagnosed with autism without intellectual disability score high in AQ and low in EQ, having a mean AQ of 35.8 compared to the control group (AQ= 16.4) and a mean EQ of 20.4 compared to 42.1 for the control group [10].

The AAA battery was developed on the British population, with the first validation study for AQ published in 2001 [8], followed by the validation of EQ in 2004 [11]. Subsequently, multiple validation studies were conducted in Canada on the French-speaking population [17], Australia [18], and The Netherlands [19]. These studies produced results consistent with the British validation study with similar findings regarding reliability and validity while confirming the negative correlation between EQ and AQ.

Another important concept for ASD is alexithymia, which is the difficulty to recognize and process emotions. It is common in people with ASD due to their differences in sensory perception, social comprehension, and external experience focusing [20], which is why we used PAQ (Perth Alexithymia Questionnaire) and BVAQ (Bermond-Vorst Alexithymia Questionnaire), two scales measuring alexithymia, in order to assess the construct validity of the AQ. We also decided to include a concurrent scale for EQ, the QCAE (Questionnaire of Cognitive and Affective Empathy), due to the lack of a validated scale for autistic traits and empathy in the Romanian population.

The main difficulty with applying the AQ and EQ in the general population is the differential diagnosis, as many individuals with personality disorders (especially cluster A) score

highly on the AQ and lower on the EQ [21], further emphasizing the fact that all questionnaires are very useful for screening, but the diagnosis must be formulated in association with the clinical presentation, personal history, and correspondence to diagnostic criteria.

While AQ and EQ have been validated in several countries and languages, no study up to this date has assessed the psychometrics of these two scales in the Romanian population. The aim of our study was to translate and assess the psychometric qualities of AQ and EQ in a Romanian general population and to perform construct validity analysis of AQ and EQ. In the original validation study, the EQ had high consistency and validity because it took into consideration both the affective and cognitive dimensions of empathy, with a high test-retest validity (0.97) and an alpha Cronbach of 0.92 [11].

The Present Study

The main purpose of this study was to examine the psychometric proprieties in a Romanian population of two screening instruments used for identifying autism without intellectual disability in the general population, namely AQ and EQ. Before testing the psychometric properties of the instruments, we first examined potential differences in the AQ and EQ scores based on several demographic characteristics. For the psychometric evaluation of the AQ and EQ, we assessed the reliability of the instruments, focusing on internal consistency and test-retest stability. We also evaluated construct validity by examining both convergent and divergent validity through correlations with other instruments, and we assessed structural validity by analyzing the factorial structure. Due to the limited addressability of the clinical population and the small number of organizations or specialists working with autistic adults without intellectual disabilities, we decided to focus our validation solely on the general population.

Method

Translation

The translation to Romanian of the AQ and EQ scales of the AAA battery was supervised by a board of 3 specialists in mental health with an elevated level of knowledge related to the autism terminology in both Romanian and English. Two external translators specialized in mental health made two concurrent translations from English to Romanian, which were then compared by the three members of the board, resulting in a first Romanian translation unanimously accepted by each member. Subsequently, this version was sent to an authorized translator not specialized in mental health, who translated it back into English. This second English version, together with the original English version of both scales, were sent to another authorized translator not specialized in mental health who underlined the differences between the translated version and the original and stated that there are no notable differences.

The first Romanian translation, which was unanimously accepted by the 3 members of the board, was then sent to a small sample of 10 people from the general population with an education level varying from middle school to master's degree who offered feedback

regarding their ability to understand each item. All feedback was centralized by the board and the final adjustments were made.

Measures

In addition to the AQ and EQ, which were described above, three concurrent scales (in the process of validation on the Romanian population at the moment of data collection) were used for supplementary analyses.

PAQ is a self-report instrument that evaluates alexithymia in adults and adolescents. It contains 24 items that are measured on a 7-point Likert scale, where 1 is “strongly disagree” and 7 is “strongly agree”. The instrument supposedly measures five factors that are linked to alexithymia (e.g., difficulty in identifying positive and negative feelings) [22].

BVAQ consists of an extended version of a 40-item scale composed of two parallel scales, A and B. The items are scored on a 5-point Likert scale, where 1 is “definitely applies to me” and 5 is “in no way applies to me” [23].

QCAE is a scale for empathy in adults. It is composed of 31 items on a 4-point Likert scale, where 1 is “strongly agree” and 4 is “strongly disagree”, with separate scores for Cognitive and Affective Empathy [24].

The above-mentioned scales were validated on the Romanian population at the West University of Timisoara but were not published (Babotan, A., 2022, unpublished dissertation thesis; Florescu, E., 2022, unpublished dissertation thesis).

Participants and Procedure

Our first sample had a total of $N=916$ participants, of which the majority were females (693, 75.65%). The mean age of the sample was $M=32.10$, $SD=11.61$ ($Min=18$, $Max=82$). The majority of our participants were urban residents (647, 70.63%). In terms of education, most of our participants completed high school (378, 41.12%), followed by participants who had undergraduate studies (296, 32.31%) and graduate studies (205, 22.37%).

A total of 329 participants (32.91%) reported that they benefited from psychological services at least once. Likewise, 129 (14.08%) participants were consulted at least once by a psychiatrist and 37 (4.03%) by a speech therapist. Six participants (0.65%) reported having a personal diagnosis of AS, while 34 (3.71%) indicated the presence of this diagnosis in their families.

Two years after the initial data collection, a follow-up was conducted by inviting all 916 participants to fill out a shorter version of the questionnaire containing only AQ and EQ. Of these, 108 participants responded.

The study was approved by the ethics committee of the Carol Davila University of Medicine and Pharmacy Bucharest, Doctoral School (No. 36286/21.12.2021), and by the ethics committee of the Alexandru Obregia Hospital in Bucharest, Romania (No. 239/04.01.2022).

A Google Forms questionnaire was constructed and then distributed on social media platforms (Facebook, WhatsApp, Instagram, e-mail) to people in the general population. This method of distribution was preferred in order to ensure a diversity of participants, as populations from other validation studies were mainly recruited from universities, and the participants had a small age gap. The anonymized dataset can be found in the OSF data repository at the following link: <https://doi.org/10.17605/OSF.IO/WV93G>.

The form had an estimated 30-minute completion time containing 215 items included in the following categories: informed consent, confidentiality agreement, and a unique identification code (used later in the test-retest phase, code consisting in a combination of letters and numbers: date of birth, initials of the participants, and initials of the participant's mother); demographics (age, gender, living environment, education level and level of contact with mental health services- psychologist, psychiatrist, speech therapist); AQ 50 items, Eq. 60 items, PAQ (24 items), BVAQ (40 items) and QCAE (31 items). PAQ, BVAQ, and QCAE are scales with similar-construct or opposed-construct with AQ and EQ and were used to assess the construct validity of AQ and EQ.

Analytic Strategy

Before testing the psychometric proprieties of the AQ and EQ, we examined in an exploratory manner differences in our target instruments between different categories included in our sample (e.g., AQ and EQ differences between participants with different levels of education). This approach allowed us to identify potential demographic moderators of high-functioning autism indicators in the general population.

To test the instruments' reliability, we examined their internal consistency. First, we computed Cronbach's alpha indicator for AQ and EQ as unidimensional instruments, then for each of the five subscales of AQ. The standard thresholds were used for interpretation: $\alpha > 0.70$ was deemed acceptable, while $\alpha > 0.80$ was regarded as good internal consistency [25].

Next, we tested the stability of the results over time by retesting a subset of our sample after a 2-year period. This was accomplished by computing the Spearman's rank correlation coefficient (the non-parametric test was chosen because the AQ/EQ scores were not normally distributed in the test and retest samples) between the initial scores for the target instruments and the scores obtained from the retest. Selecting the subset involved notifying and inviting the participants from the initial sample. The final subset comprised the participants who agreed to participate in the retest part of the study.

To assess the validity of the AQ and EQ, we first tested the construct validity. This was accomplished by correlating the AQ and EQ scores with other instruments that measure similar (convergent validity) or opposite concepts (divergent validity). Therefore, we expected AQ to be convergent with measures of alexithymia (PAQ) and divergent with measures of empathy (QCAE). For EQ, we expected the opposite. Finally, we tested the factorial structure of the AQ to assess its structural validity. To do so, we used a confirmatory factor analysis (CFA) procedure using the diagonally weighted least squares (DWLS) method for estimating. We used CFA because there is already a strong body of literature indicating a 5-factor solution for the AQ. Hence, we wanted to confirm it for our Romanian sample [26]. Likewise, the DWLS method for estimating is recommended for ordinal or dichotomous data, especially if the normality of the distribution assumption is not met [27].

Several fit indices were important for evaluating the goodness of fit of our models: the comparative fit index (CFI), the root mean square error of approximation (RMSEA), and the standardized root mean squared residual (SRMR). We also reported the Tucker-Lewis index (TLI). To interpret these indices, we followed the guidelines established by Hu and Bentler [28] and Browne and Cudeck [29]. According to these guidelines, CFI and TLI

values should be close to 0.95 or higher to indicate a good fit. RMSEA and SRMR values below 0.08 also suggest a good fit.

Results

Sample Description and Exploratory Analyses

As already mentioned, our sample was mainly comprised of female participants. The mean scores of AQ and EQ relative to the participants' gender are presented in Table 1.

Because our sample failed to meet the criteria for the normality of distribution, we tested on a non-parametrical level to see whether the differences between different categories on the AQ and EQ measures were statistically significant.

The Kruskal-Wallis test revealed significant differences among gender categories with respect to AQ and EQ scores, as well as educational categories.

Wilcoxon rank-sum test was used to explore the differences between the categorical variables that only had two groups; the results revealed that, regarding area, people coming from the urban area had statistically significantly different scores on AQ compared to people coming from the rural area. This was not the case for EQ, where no significant differences were found among the area categories.

Participants who were referred to a psychologist at least once scored significantly different on EQ compared to participants who were not. Similar results were found between participants who were referred to a psychiatrist or speech therapist at least once and those who were not for both AQ and EQ. Table 2 presents data for AQ and EQ with respect to educational levels and psychologist referrals.

For the AQ and EQ scores relative to the educational level, we also computed the post-hoc tests. We ran Wilcoxon rank-sum tests to capture specific differences between educational categories. Because multiple comparisons were involved, we used the Bonferroni correction to adjust the significance threshold (standard $\alpha=0.05$ divided by the number of comparisons: $0.05/15=0.003$). Regarding the AQ score, it was revealed that there are significant differences between high school and undergraduate ($Z = -3.72, p<.001$), high school and graduate ($Z = -7.82, p<.001$), high school and doctorate ($Z = -3.29, p<.001$), and undergraduate and graduate levels ($Z = -4.61, p<.001$). Regarding the EQ score, there were significant differences between the high school and graduate ($Z = -5.13, p<.001$), high school and doctorate ($Z = -3.72, p<.001$), undergraduate and graduate ($Z = -0.3.03, p=.002$), and undergraduate and doctorate levels ($Z = -3.13, p=.002$).

In the original study by Baron-Cohen et al. [8], a cut-off score of 32 or higher on the AQ was proposed. Among our participants, 19 individuals (1.9%) met this criterion. All partici-

Table 1 AQ / EQ scores split by gender

Gender / Scale score	Males, N=217 (23.68%)	Females, N=693 (75.65%)	Nonbi-nary, N=5 (0.54%)	Ques-tioning, N=1 (0.10%)
AQ	18.92 (5.75) ¹	17.89 (6.26)	24.20 (9.58)	32 (NA)
EQ	42.35 (11.93)	49.28 (12.16)	35.60 (12.24)	20 (NA)

¹ Mean (SD)

Table 2 Medians (interquartile ranges) and significance tests for AQ and EQ on different educational levels and psychologist referrals

Psychologist visit / Scale score		No, $N=587^1$		Yes, $N=329^1$	p -value ²		
AQ		18.0 (7.0)		18.0 (9)	0.40		
EQ		48 (17)		50 (18)	0.02		
Education / Scale score	Middle school $N=8^1$	High school $N=378^1$	Under-graduate $N=296^1$	Graduate $N=205^1$	Doctoral $N=22^1$	Post-doc-toral $N=7^1$	p -value ³
AQ	20.0 (4.5)	19.0 (7.0)	17.0 (8.0)	15.0 (8.0)	13.5 (9.0)	13.0 (10.0)	<0.001
EQ	43 (7)	47 (18)	49 (18)	53 (15)	58 (16)	50 (19)	<0.001

¹ Median (IQR), ² Wilcoxon rank-sum test, ³ Kruskal-Wallis rank-sum test

Table 3 Internal consistency measures (Cronbach's α) in the original study and in our sample for AQ

Sample	AQ	AQ social skill	AQ attention switching	AQ attention to detail	AQ communication	AQ imagination
Original	-	0.77	0.67	0.63	0.65	0.65
Present	0.75	0.58	0.62	0.60	0.50	0.40

pants who reported having an ASD diagnosis ($N=6$) were in this group. Additionally, of those who scored above 32 on the AQ, 16 participants indicated that they had used psychological services at least once, while 13 participants reported having consulted a psychiatrist at least once.

According to the cut-off score proposed by Baron-Cohen and Wheelwright [11] for the EQ, a score of 30 points or lower identifies individuals with HFA. In our study, 93 participants, representing 10.2%, fell within this category. Among the six participants who reported having a diagnosis of ASD, only four scored 30 points or less. Furthermore, only a portion of these individuals sought help, with 26 participants benefiting from psychological services and 15 participants receiving psychiatric consultations.

Reliability Measures

Internal Consistency

Unlike the original validation study [8], our sample had lower internal consistency coefficients for the AQ scale. While the whole instrument presented an acceptable internal consistency index ($\alpha=0.75$), only poor to moderate coefficients were obtained when the subscales were considered. Compared data are presented in Table 3.

While the AQ obtained modest internal consistency coefficients, the EQ performed significantly better. It obtained an $\alpha=0.88$, similar to the coefficient reported in the original study ($\alpha=0.92$) [11].

Test-Retest Correlation

The reliability of the two instruments was further analyzed by correlating the scores obtained in the initial evaluation phase with those obtained from a smaller sample ($N=108$) after two years. Since the distributions of the four variables (test and retest measures for AQ; test and retest measures for EQ) did not meet the assumptions of normality, we opted for the non-parametric Spearman correlation test. For AQ, we obtained a moderately high positive correlation, $r(106)=0.72, p<.001$. This coefficient was similar to the one obtained in the original study [8, 11] ($r=.7, p<.05$).

Similarly, for EQ, a moderately high positive correlation was indicated, $r(106)=0.79, p<.001$. However, in this case, the original study [8, 11] presented a much higher correlation index after a 12-month period between the assessment phases ($r=.97, p<.001$).

Validity Measures

Construct Validity

The construct validity for AQ and EQ was assessed by correlating the total scores of the scales with those of similar-construct or opposed-construct instruments. AQ was positively correlated with PAQ and negatively correlated with QCAE, questionnaires that measure alexithymia and cognitive and affective empathy. As expected, it negatively correlated with EQ. Likewise, EQ positively correlated with instruments that measure similar concepts (QCAE) and negatively correlated with alexithymia (PAQ). Correlational data are presented in Table 4.

Structural Validity

Regarding the structure of the scale, because EQ is reported to be unidimensional [11], we only focused on the five subscales of the AQ. The results of the CFA indicated that 4 subscales were highly correlated– social skills, communication, attention switching, and imagination (correlation coefficients ranged from $r = -.55$ to -0.86), and the remaining attention to detail was small to moderately correlated to the other subscales. These results indicated an overlap between the factors, and this overlap was confirmed by the poor fit indices obtained in the 5-factor model.

Studies that faced similar issues in their samples with the AQ structure proposed a hierarchical factor model, with one overarching (second-order) social factor encompassing the 4 highly correlated domains and the other one a separate factor including the attention to detail [19]; others reduced the structure of the scale by eliminating items, thus resulting a

Table 4 Correlation matrix for AQ, EQ, PAQ, BVAQ and QCAE

	AQ	EQ	PAQ	BVAQ	QCAE
AQ					
EQ	-0.498***				
PAQ	0.517***	-0.453***			
BVAQ	-0.361***	0.527***	-0.661***		
QCAE	-0.330***	0.733***	-0.325***	0.548***	

Computed correlation used spearman-method with listwise-deletion
*** $p<.001$

Table 5 Fit indices for the structural validity of the AQ

Model	CFI	TLI	SRMR	RMSEA
5-factor (original)	0.58	0.56	0.11	0.05
2-factor	0.56	0.54	0.12	0.05
5-factor (post-hoc modification)	0.83	0.81	0.09	0.04

Table 6 Internal consistency (Cronbach's α): original vs. trimmed model

	General	Social skill	Attention Switching	Attention to Details	Communication	Imagination
Original model	$\alpha=0.75$	$\alpha=0.58$	$\alpha=0.62$	$\alpha=0.60$	$\alpha=0.50$	$\alpha=0.40$
Post-hoc modified model	$\alpha=0.79$	$\alpha=0.59$	$\alpha=0.63$	$\alpha=0.63$	$\alpha=0.52$	$\alpha=0.43$

Table 7 Fit indices for EFA of AQ

Factor	Chi-square	df	CFI	RMSEA
1	4990.09	1175	0.542	0.060
2	3284.81	1126	0.741	0.046
3	2243.76	1078	0.860	0.034
4	1735.29	1031	0.916	0.027
5	1472.67	985	0.941	0.023

4-factor solution with 43 items [30] and a 3-factor solution with 26 items [31]. However, the solution that provided the most satisfactory fit indices proposed the retention of the 5-factor model but adjusted it by eliminating the items that failed to load on the factors with at least 0.40 [32].

Based on these considerations, we tested using CFA for the original 5-factor solution and the 2-factor solution (based on Hoekstra et al., 2008, but only with first-order factors) [19]. Regarding the factor loadings, 16 items failed to load more than 0.40 on their factors (from social skill item 13; from attention switching items 16 and 43; from attention to detail items 12, 28, 29, 30 and 49; from communication items 18 and 38; from imagination items 14, 21, 24, 40, 41 and 42). Because the initial fit indices were not near the recommended thresholds, we decided to eliminate post-hoc the items with poor factor loadings. The fit indices for the three models are presented in Table 5.

The internal consistency of the instrument was once again examined after we ran the factorial analysis. The post-hoc modified model presented slightly higher internal consistency compared to the original model across all subscales of the instruments as well as the entire scale. The results are presented in Table 6.

Because we were unable to confirm an adequate fit to the data for the existing models we tested, and using post-hoc methods in CFA is generally discouraged due to high chances of overfitting the model [33], we have decided to run an EFA for outlining the factorial structure of the AQ based on our sample. Using the DWLS method for estimating and the Oblimin rotation due to the binary data, a 5-factor structure of the scale presented adequate fit indices, as revealed in Table 7. Because CFI improves as more factors are added, and RMSEA decreases, a 5-factor structure is recommended based on our data set.

The five identified factors demonstrated moderate correlations ranging from 0.1 to 0.3, indicating that while they are related to a moderate degree, they are distinct factors that capture specific variability in the individuals' responses.

However, the factorial structure returned by the EFA was significantly different compared to the original one. After eliminating the items that failed to load more than 0.40 on their factors, the factorial structure retrieved from EFA was the following: for factor 1– items 5, 6, 7, 9, 12, 16, 19, 23, 29, 41; for factor 2– items 3, 8, 10, 27, 31, 36; for factor 3– items 17, 38; for factor 4– items 20, 35, 45; for factor 5– items 11, 15, 22, 25, 26, 34, 37, 44, 46, 47.

Out of the factors analyzed, only the fifth and first factors closely resemble the original structure of the AQ. The items loading on the fifth factor relate to social skills, while most items on the first factor pertain to attention to detail. The other three domains were not sufficiently represented in the original factorial structure of the AQ. Specifically, factor 2 suggested a combination of the original domains of Imagination, Attention Switching, and Communication, while the other two factors did not have enough items with strong loadings.

Regarding the internal consistency of the factors, only these two factors corresponding to social skills and attention to detail exceeded the optimal threshold of 0.70 for Cronbach's alpha ($\alpha=0.78-0.70$).

Based on these findings, we decided to run another CFA that only takes into account these two factors. This last model returned significantly improved fit indices, CFI=0.92, TLI=0.91, RMSEA=0.05, SRMS=0.09. A single item failed to load more than 0.40 on its factor, item 29. The two factors were moderately correlated with each other, 0.29, indicating that they represent distinct domains.

Discussion

The current study investigated the psychometric properties of the AQ and the EQ in a large sample retrieved from the general Romanian population. Our analysis aimed to assess the reliability of these instruments, focusing on both internal consistency and test-retest stability, as well as their validity, including construct and factorial structure. Regarding internal consistency, AQ presented an acceptable overall internal consistency index, but when the subscales were considered, lower indices were obtained. This means that, for the Romanian population, some of the items that form the subscales of the AQ are not highly representative of that specific construct (e.g., for the imagination subscale, the items might not be very representative for the construct, hence the poor inter-relatedness between items). Even so, the overall instrument proved to be reliable, which indicates that, on a global score, the instrument may provide similar results when testing autism without intellectual disability in the Romanian population over time.

Even though the internal consistency indices for the subscales of the AQ were unsatisfactory, these results were not contrary to the performance of the instrument in other populations. As already mentioned, except for the social skill subscale, all other subscales obtained $\alpha < 0.70$ in the original study [8]. Similarly, in the Australian population, for the typically developed sample, the internal consistency measures provided the same acceptable total index ($\alpha=0.75$) but poor coefficients when subscales were considered [18]. The French-Canadian version of the scale yielded, once again, similar results [17], and so did the Swiss adaptation [34].

With respect to EQ, the same analysis revealed a higher internal consistency index, meaning that the instrument is highly reliable in the stability of the results in the Romanian population. The fact that the two instruments are overall reliable and stable over time was

further demonstrated by the satisfactory test-retest correlation indices provided. The reliability of the scales is further emphasized, considering that the period between the two evaluations was two years.

The construct validity of the AQ was indicated by the positive association with an instrument measuring alexithymia and negative association by an instrument measuring empathy. Similar arguments were provided for EQ, indicating that it resembles a valid instrument with respect to capturing empathy in the target sample.

Regarding the structural validity of the AQ instrument, our study confirms on Romanian population what previous studies have indicated, meaning that it uses more items to measure autism without intellectual disability with its facets than is needed. This was demonstrated by the multiple factorial models tested, with the final one comprised of a 5-factor adjusted model (by eliminating items poorly loaded on their factors) providing the most satisfactory fit indices.

Out of 916 participants, 6 reported having a diagnosis of AS, 4 females and 2 males, contrary to the gender expectation [35], which can be explained by the bigger number of females in our sample, but since the number of previously diagnosed people is so small in our sample, no general conclusion can be drawn. All 6 declared they had a psychotherapy process and had a score above the cut-off-scores for AQ (32+), but only 4 of them correlated with cut-off-scores for EQ (below 30) from the English version [8, 10]. The higher EQ scores of the two participants could be attributed to the therapeutic process they have undergone. The 0.665% of the population is in the range of the expected rate [5, 6]. Even though the diagnosis was not clinically confirmed by the research team, given the large sample included in this study, a prevalence of 0.665% for autism without intellectual disability in the Romanian general population can be speculated, given the present data. Regarding the cut-off score identified in other countries, the authors report the following cut-off scores: AQ=29 for the Australian population [18], AQ=26 for the French population [36], and AQ=33 for the French-Canadian population [17]. All these cut-off values effectively encapsulate the participants who reported having a diagnosis of ASD in our sample; however, an optimal cut-off score should be established based on the comparison between the general and clinical samples.

Although we managed to prove that AQ and EQ have good internal consistency and validity in a sample of the Romanian general population, one of the limitations is the fact we did not include a group of clinical population in order to establish cut-off values for both scores and the RQ questionnaire from the AAA battery was also not included. Another limitation is the fact that the three scales used for concurrent validation were translated into Romanian, but the translation was not officially validated at the moment of data collection for this study.

Conclusions

The Romanian translations of AQ and EQ are reliable instruments for measuring autistic traits in a general population, as we obtained similar results with the English version [8] on a significant sample size (916 participants).

Even though the Romanian version of AQ presents slightly lower structural validity than the original study, while the structural validity of EQ is similar, both instruments present

a valuable resource for clinicians working with neurodivergent patients, as the interest in this area of pathology has increased greatly in Romania over the past few years. The open-source feature, along with a strong body of literature testing the instruments and previous optimization attempts for specific populations, offers several advantages for using these instruments as screening tools for ASD in adults without cognitive and/or language delay. Future studies that include clinical population are needed to determine a cut-off score in order to use these instruments as screening tools for ASD.

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Declarations

Ethical Approval This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee of Carol Davila University of Medicine and Pharmacy, Bucharest, Romania, No. 36286/ Date 12/21/2021 and by the Ethics Committee of Alexandru Obregia Hospital, Bucharest, Romania, No. 239/ Date 01/04/2022.

Consent to Participate Informed consent was obtained from all individual participants included in the study. A brief description of the methodology and purpose of the study followed by an informed consent were attached at the beginning of the Google Forms questionnaire sent to each participant both in the initial and follow-up phase. All data was collected anonymously as each participant was assigned a unique code. All the participants also offered their informed consent in advance for the eventual publication of an article regarding the data collected.

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